Project Documentation and Links to the Sources

1 Business Area From POI Clusters (Geometry Pipeline)

This section defines a geometric business area using business-related POIs. The procedure is: (i) project coordinates to a metric CRS, (ii) restrict to a core analysis region, (iii) collect business POIs, (iv) cluster with DBSCAN, and (v) convert the dominant cluster to a polygon. The polygon area is the business area, and its centroid is the business center.

Coordinate system and core region

We work with a city boundary polygon B and two reference points: a geocoded city center c and a historic center g, both initially in lon-lat.

$$\textbf{Inputs: } B \subset \mathbb{R}^2,\ c = (\lambda_c, \varphi_c),\ g = (\lambda_g, \varphi_g),\ R \text{ (km)}$$

A UTM zone is chosen from longitude, and the EPSG code is determined by hemisphere:

$$z = \left\lfloor \frac{\lambda + 180}{6} \right\rfloor + 1$$
$$\text{EPSG} = \begin{cases} 32600 + z, & \varphi \geq 0, \\ 32700 + z, & \varphi < 0 \end{cases}$$

Let T be the projection from lon-lat into the chosen local metric CRS:

$$T:\ (\lambda, \varphi) \mapsto (x, y) \in \mathbb{R}^2$$

We define a circular core region of radius $1000R$ meters around the projected city center, and intersect it with the projected city boundary:

$$C = \left\{ u \in \mathbb{R}^2 :\ \|u - T(c)\| \leq 1000R \right\}$$
$$K = T(B) \cap C$$

If the boundary intersection fails (empty or zero area), we fall back to the core circle:

$$K = \begin{cases} K, & \text{Area}(K) > 0, \\ C, & \text{Area}(K) = 0 \end{cases}$$

POI clustering and polygon construction

Let $\mathcal{S}$ be the set of OSM tags considered "business". The metric POI point set is:

$$X = \left\{ T(p)\ :\ p \in \text{OSM},\ p \in T^{-1}(K),\ \text{tag}(p) \in \mathcal{S} \right\}$$

DBSCAN is applied to X using radius ε and minimum samples m:

$$N_\varepsilon(x_i) = \{x_j \in X :\ \|x_j - x_i\| \leq \varepsilon\}$$

$$x_i \text{ is core} \iff |N_\varepsilon(x_i)| \geq m$$

DBSCAN assigns labels:

$$\ell_i \in \{-1, 0, 1, 2, \dots\}, \qquad (\ell_i = -1 \text{ means noise})$$

We keep the dominant (largest) non-noise cluster:

$$k^\star = \arg\max_{k \geq 0} |\{i : \ell_i = k\}|$$
$$X^\star = \{x_i \in X : \ \ell_i = k^\star\}$$

To obtain a continuous footprint, buffer each cluster point by radius r and take the union:

$$\mathcal{B}(x, r) = \left\{u \in \mathbb{R}^2 : \ \|u - x\| \leq r\right\}$$
$$U = \bigcup_{x \in X^\star} \mathcal{B}(x, r)$$

If U yields multiple polygons, keep the largest component:

$$P = \text{LargestPolygon}(U)$$

Optionally simplify with tolerance s:

$$P_s = \begin{cases} \text{Simplify}(P, s), & s > 0, \\ P, & s \leq 0 \end{cases}$$

Business area and business center:

$$A_{\text{km}^2} = \frac{\text{Area}(P_s)}{10^6}$$
$$\mu = \text{Centroid}(P_s)$$

Fallback if no polygon is available:

$$\mu = T(c), \qquad A_{\text{km}^2} = 0$$

Historic-to-business distance (in meters):

$$g_m = T(g)$$
$$d = \|\mu - g_m\| = \sqrt{(\mu_x - g_x)^2 + (\mu_y - g_y)^2}$$

2 Model-Based Shares and Fixed-Point Business Center

This section defines a model that assigns each grid cell a business "share" in $[0, 1]$ via a logistic specification, and then computes an economic center μ that is consistent with those shares. Consistency is required because travel time depends on μ, while μ is a weighted average of cell locations using the shares.

Cells, features, and standardization

Let grid cells be indexed by $i = 1, \ldots, N$, with metric centroids $c_i = (c_{x,i}, c_{y,i}) \in \mathbb{R}^2$. Each cell has observed features:

$$d_i^{(g)},\ h_i,\ p_i,\ \ell_i,\ r_i$$

To avoid line overflow, the mapping to data columns is written on multiple lines:

$$\begin{aligned} d_i^{(g)} &= \texttt{dist_to_g_m}, & h_i &= \texttt{herit_density_500m}, \\ p_i &= \texttt{pop_density_per_km2}, & \ell_i &= \texttt{log_rent}, \\ r_i &= \texttt{rent_missing}. \end{aligned}$$

Two interactive multipliers are applied: transport factor t and historic-amenity strength a:

$$t \in [0.1, 2.0], \qquad a \in [0.1, 2.0]$$

Let v be travel speed in meters per minute:

$$v = \text{speed_m_per_min}$$

Each feature is standardized using training means and standard deviations:

$$z_x(i) = \frac{x_i - \mu_x}{\sigma_x}, \qquad \sigma_x \neq 0$$

Travel time depends on the unknown center

For a candidate business center $\mu = (\mu_x, \mu_y)$, define distance and travel time:

$$\begin{aligned} d_i(\mu) &= \|c_i - \mu\| = \sqrt{(c_{x,i} - \mu_x)^2 + (c_{y,i} - \mu_y)^2} \\ \tau_i(\mu, t) &= \frac{d_i(\mu)}{v\, t} \end{aligned}$$

Logit model for shares

Define standardized regressors:

$$\begin{aligned} z_\tau(i; \mu, t) &= \frac{\tau_i(\mu, t) - \mu_\tau}{\sigma_\tau} \\ z_g(i) &= \frac{d_i^{(g)} - \mu_g}{\sigma_g}, \qquad z_h(i) = \frac{h_i - \mu_h}{\sigma_h} \\ z_p(i) &= \frac{p_i - \mu_p}{\sigma_p}, \qquad z_\ell(i) = \frac{\ell_i - \mu_\ell}{\sigma_\ell}, \qquad z_r(i) = \frac{r_i - \mu_r}{\sigma_r} \end{aligned}$$

Linear predictor:

$$\begin{aligned} \eta_i(\mu, t, a) = \beta_0 &+ \beta_\tau z_\tau(i; \mu, t) + \beta_g\,(a\, z_g(i)) + \beta_h\,(a\, z_h(i)) \\ &+ \beta_p z_p(i) + \beta_\ell z_\ell(i) + \beta_r z_r(i) \end{aligned}$$

Numerical clamp:

$$\tilde{\eta}_i(\mu, t, a) = \min\{35, \max\{-35, \eta_i(\mu, t, a)\}\}$$

Share (logistic):

$$s_i(\mu, t, a) = \frac{1}{1 + \exp\bigl(-\tilde{\eta}_i(\mu, t, a)\bigr)}$$

Fixed-point definition of the business center

Given shares, the implied business center is the weighted average of cell centroids:

$$S(\mu, t, a) = \sum_{i=1}^{N} s_i(\mu, t, a)$$

$$F(\mu; t, a) = \frac{1}{S(\mu, t, a)} \sum_{i=1}^{N} s_i(\mu, t, a)\, c_i$$

A fixed point $\mu^\star$ satisfies $\mu^\star = F(\mu^\star; t, a)$. We approximate it iteratively from $\mu^{(0)} = (\mu_{0x}, \mu_{0y})$:

$$\mu^{(k+1)} = F(\mu^{(k)}; t, a)$$

$$\left\| \mu^{(k+1)} - \mu^{(k)} \right\| < \delta \quad \text{(stop)}$$

Outputs:

$$\mu^\star = \mu^{(K)}, \qquad s_i^\star = s_i(\mu^\star, t, a)$$

Separation from the historic center $g = (g_x, g_y)$:

$$\mathrm{sep}_m = \|\mu^\star - g\|$$

Quantization (for compact browser transfer):

$$b_i = \mathrm{round}(255 \cdot \min\{1, \max\{0, s_i^\star\}\}) \in \{0, \ldots, 255\}$$

3 Interactive Dashboard: Linear-City Theoretical model

This section documents the interactive dashboard. The goal of the dashboard is to visualize, in real time, the equilibrium business segment and the economic center in a one-dimensional (linear) city model. The model and the closed-form equilibrium formulas follow the paper [1].

State variables and derived quantities

The city is the interval $[-1, 1]$. Agents occupy land with unit density, and total land length is 2. The slider `N` represents the mass of residents; the implied mass of firms is:

$$M = 2 - N, \qquad 0 < N < 2$$

The historic center location is `g`:

$$g \in [0, 1]$$

The model uses two cost parameters: transport cost $t \geq 0$ and historic amenity strength $e > 0$. Their ratio is the key dimensionless quantity:

$$\theta = \frac{t}{e}$$

The business area is an interval $[q, p]$ with length M, and the economic center is its midpoint:

$$p - q = M, \qquad \mu = \frac{p + q}{2}$$

The model enforces feasibility in the city boundaries:

$$-1 \leq q < p \leq 1$$

Closed-form candidate equilibria (three cases)

The equilibria computes three closed-form candidates, corresponding to the relative position of g and $[q, p]$. These match the three cases in the paper [1].

Case A (historic center inside business area). Assume

$$q \leq g \leq p.$$

Define the candidate economic center and boundaries:

$$\mu_A = \frac{Ng}{N - 2\theta}$$
$$p_A = \mu_A + \frac{M}{2}, \qquad q_A = \mu_A - \frac{M}{2}.$$

Case B (historic center to the right of business area). Assume

$$q < p \leq g.$$

For $\theta > 0$, the candidate is independent of g:

$$\mu_B = -\frac{MN}{4\theta}$$
$$p_B = \mu_B + \frac{M}{2}, \qquad q_B = \mu_B - \frac{M}{2}.$$

Case C (historic center to the left of business area). Assume

$$g \le q < p.$$

For $\theta > 0$, the symmetric candidate is:

$$\mu_C = \frac{MN}{4\theta}$$
$$p_C = \mu_C + \frac{M}{2}, \qquad q_C = \mu_C - \frac{M}{2}.$$

Case selection rule

The model computes all available candidates and then selects the one that best satisfies the case constraints and city feasibility constraints. Conceptually, it chooses the candidate with the smallest constraint violation.

Let viol$(\cdot)$ measure violations of (i) city bounds and (ii) case assumptions. One convenient form is:

$$\text{viol} = \underbrace{\max(0, -1-q) + \max(0, p-1)}_{\text{out of city}} + \underbrace{\text{viol}_{\text{case}}}_{\text{case mismatch}},$$

where

$$\text{viol}_{\text{case}} = \begin{cases} \max(0, q-g) + \max(0, g-p), & \text{Case A}, \\ \max(0, p-g), & \text{Case B}, \\ \max(0, g-q), & \text{Case C}. \end{cases}$$

What each plot shows

Plot 1: City layout (instant response). It displays the city segment $[-1, 1]$, the business interval $[q, p]$, the historic center g (marker), and the economic center μ (marker). The plotted geometry is:

$$\text{city: } [-1, 1], \qquad \text{business: } [q, p], \qquad \text{markers: } g,\ \mu.$$

Plot 2: Sweep historic center location (Figure 1 style). Holding (N, t, e) fixed, it sweeps g on a grid and recomputes $(q(g), p(g))$:

$$g_k = \frac{k}{K-1} \cdot g_{\max}, \qquad k = 0, \ldots, K-1,$$

$$(q_k, p_k, \mu_k) = \text{SolveBounds}(N, t, e, g_k).$$

It then draws curves $p(g)$ and $q(g)$ and overlays the current slider value g as a dashed reference line.

Plot 3: Sweep transport cost (Figure 2-3 style). Holding (N, g, e) fixed, it sweeps t and recomputes $(q(t), p(t))$:

$$t_k = t_{\min} + \frac{k}{K-1}(t_{\max} - t_{\min}), \qquad k = 0, \ldots, K-1,$$

$$(q_k, p_k, \mu_k) = \text{SolveBounds}(N, t_k, e, g).$$

It draws $p(t)$ and $q(t)$, overlays the fixed g as a dashed vertical line, and marks the current slider value t.

4 External Resources and Links

Data sources (datasets)

- WorldPop (Armenia 2020 population raster, 100m): `https://data.worldpop.org/GIS/Population/Individual_countries/ARM/Armenia_100m_Population/ARM_ppp_v2c_2020.tif`
- OpenStreetMap (data used via OSMnx): `https://www.openstreetmap.org/`
- Nominatim (geocoding service): `https://nominatim.org/`
- Nominatim usage policy: `https://operations.osmfoundation.org/policies/nominatim/`
- Overpass API (public endpoint): `https://overpass-api.de/api/interpreter`
- Kaggle datasets:
  - `https://www.kaggle.com/datasets/molchoun/real-estate-data-armenia`
  - `https://www.kaggle.com/datasets/mirzoyanvahe/armenian-real-estate-market-data`
  - `https://www.kaggle.com/datasets/puff1n/armenian-flats-from-listam`

Map tile source (Leaflet)

- CARTO raster tiles (Positron / light_all): `https://\{s\}.basemaps.cartocdn.com/light_all/\{z\}/\{x\}/\{y\}\{r\}.png`

Other external links (libraries/assets)

- Leaflet CSS: `https://unpkg.com/leaflet@1.9.4/dist/leaflet.css`
- Leaflet JS: `https://unpkg.com/leaflet@1.9.4/dist/leaflet.js`
- Plotly: `https://cdn.plot.ly/plotly-2.30.0.min.js`
- Proj4js: `https://cdnjs.cloudflare.com/ajax/libs/proj4js/2.9.0/proj4.js`
- Pako: `https://cdn.jsdelivr.net/npm/pako@2.1.0/dist/pako.min.js`
- Scrollama: `https://unpkg.com/scrollama`
- Google Fonts: `https://fonts.googleapis.com` , `https://fonts.gstatic.com`
- SVG namespace reference: `http://www.w3.org/2000/svg`
- GitHub API: `https://api.github.com`

Thesis reference and special thanks

If this document is distributed together with the thesis PDF in the same folder, the following link should open it:

Goncharov (2021) thesis PDF

Special thanks to Nikita Goncharov for the theoretical framework on asymmetric city structure and the separation of historic and economic centers, which motivates the full project and proposes a theoretical background.

References

[1] N. Goncharov, *Asymmetric city: spatial separation of historic and economic city centers*, Master Thesis, New Economic School, January 2021.